

Conclusion

The evolved strategy of *Yersinia pestis*

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Chapter 10

Conclusion: the evolved strategy of *Yersinia pestis*

This thesis began with one organism and one peculiarity. *Y. pestis* enters a mammalian host from a flea bite, permits itself to be taken up by macrophages, replicates inside them for a while, and only then reverses course and resists phagocytosis altogether. The reversal is not incidental book-keeping. It is among the sharpest differences between *Y. pestis* and the comparatively mild ancestor it descends from, and it is the sort of thing that invites a mathematical answer, because it is a choice between two strategies whose returns can in principle be written down and compared. I wanted to know whether the switch could be shown to pay, and to say in what currency it pays.

Most of the thesis is the machinery that question turned out to need. The choice is made by a small number of bacteria inside a single cell, where whether the lineage survives at all is settled by chance rather than by a mean, so the object required is a distribution and not a trajectory. Getting that distribution for a cell that grows its occupants and then ruptures took Chapter 4 and then Chapter 5; carrying it up to a population took Chapter 6; and asking what the release schedule is worth against defences that can be exhausted took Chapter 8. Along the way two questions detained the work longer than they strictly deserved, and both became chapters of their own: a constant governing subcritical branching conditioned on survival (Chapter 3), and the problem of writing down Markov chains whose rates are allowed to move (Chapter 9).

What follows is a reckoning rather than a summary. [Section 10.1](#) sets out what each chapter established and what it left standing. [Section 10.2](#) returns to the four questions posed in Chapter 1 and answers them as far as the work allows, which in two cases is not all the way. [Section 10.3](#) gives the one argument that never found a home in a chapter of its own: a small piece of reasoning about why an organism might want to be eaten early and not late. [Section 10.4](#) says what the thesis does not establish, and [section 10.5](#) closes.

10.1 Contributions, chapter by chapter

Chapter 2 assembles the objects and the methods: Markov chains in discrete and continuous time, the branching processes that matter here, and the generating-function machinery — forward and backward equations, extinction probabilities, limiting conditional means, and the method of characteristics — each developed once, on whichever process shows it most plainly. Two of its appendices are more than review. The first records how heavily survival near criticality depends on the size of the founding cohort (Section 2.A), which is the fact that almost everything downstream leans on. The second solves a family of absorption models in which particles are drawn into a compartment and may then replicate inside it; the hardest member of that family is solved here for the first time (Section 2.B).

Chapter 3 is the chapter furthest from plague, and it is kept because the question it answers is a real one and the answer is complete. For the subcritical binary Galton–Watson process the survival probability decays as $S_n \sim A(p)(2p)^n$, and that single constant also fixes the population conditioned on survival, whose mean converges to $1/A(p)$. The constant exists because the limit can be rewritten as a convergent infinite product. Rescaling the survival recursion to the logistic map identifies it with a value of the associated Koenigs linearising function, and a theorem of Becker and Bergweiler shows that function to be hypertranscendental (Theorem 3.5), which does not settle the closed-form question for $p \mapsto A(p)$ but does explain why a long search for a formula came to nothing. What remains is practical: the product, together with the near-critical asymptotic $A(p) \sim 2(1 - 2p)$, describes the constant across the whole subcritical range.

Chapter 4 introduces the process the rest of the thesis is built on. Inside one host cell each pathogen particle divides at rate λ , dies at rate μ , and triggers rupture of the whole cell at rate δ . The calculation closes because the probability of no rupture yet, $I(t)$, obeys an autonomous Riccati equation; the two roots a and b of its characteristic quadratic then carry every closed form in the thesis. Cell-fate probabilities, the mean and second moment of the intracellular load, the mean release and the first two conditional moments of the burst size all follow in that one coordinate, and several of them — the second moment expressed through I , the variances, the joint load–release moments — go beyond what the birth–death–catastrophe literature had recorded.

Chapter 5 completes the single cell in distribution. The load is geometric at every time, with a ratio that slides towards $1/a$; the quasi-stationary law is that endpoint (Theorem 5.6); and the burst size, averaged over the instant of rupture, is the same law (Theorem 5.10). That the two coincide is the most satisfying result in the thesis, because the two constructions share no step: one conditions a surviving trajectory on never being absorbed, the other averages over rupture times, and the identity turns out to hold because size-biasing telescopes the occupation integral onto the endpoint of the process’s own geometric ratio. Two probes then mark the boundary of the structure. Several founders sharing a cell leave it intact; chaining one cell’s burst into the next destroys it as soon as internal extinction is possible, and that case remains open.

Chapter 6 embeds the cell in a population. The survival and release statistics become age-structured kernels, and the resulting renewal system is exact in the mean. Three consequences follow. The standard model of within-host virus dynamics is a projection of that system, valid one exponential phase at a time (Theorem 6.3), so a fitted release rate is a property of the phase and not of the organism, and a model that already carries a hand-fitted eclipse alongside a bursting pathogen is fitting the same delay twice. No fit of the two population constants recovers the three intracellular rates (§6.5.6), so a population fit is worth doing only alongside a single-cell measurement. And at fixed total output under linear clearance the release schedule is invisible to R_0 (Theorem 6.6) while remaining visible to establishment: bursting strictly lowers the extinction probability of a small inoculum if and only if the flooding parameter $L = a(1 - b)$ exceeds one (Theorem 6.8).

Chapter 7 solves a two-type version exactly. Type 1 converts irreversibly into type 2, either type may trigger a single global catastrophe, and because the catastrophe rate depends on the whole path the natural object is an occupation-time transform rather than a terminal-count generating function. The backward system is triangular, and after linearisation the surviving Riccati equation becomes the Gauss hypergeometric equation, giving a closed formula for the finite-time probability of avoiding catastrophe (Theorem 7.4) which holds for any ordering of the type-specific rates and reduces to the two-type branching equations of Antal and Krapivsky [1] when the catastrophe rates vanish. Read biologically, the two types are early and adapted intracellular phenotypes and the catastrophe is irreversible loss of containment.

Chapter 8 asks what a burst is worth when the defence can be used up. Inside the cell, a replicator–suppressor process with a granule store that is never replenished makes a larger founding cohort disproportionately safer, because the cohort itself spends the store. Outside it,

a fixed removal budget ϑ per delivery gives the surviving release in closed form, $\mathcal{R}(\vartheta) = V_\infty a^{-\vartheta}$, and with it a reversal: against a threshold rather than a rate, the flooding criterion flips, and bursting beats matched-mean budding precisely when $L < 1$ (Proposition 8.12). The reason is a single fact about variance. Extinction probability rewards low offspring variance whereas the surplus above a threshold is convex and rewards high variance, so the sign of $L - 1$ drives both criteria and drives them opposite ways. The chapter also yields the number that reappears below: a removal budget $\vartheta^* = \log V_\infty / \log a$ beyond which no intracellular strategy stays supercritical (Corollary 8.8).

Chapter 9 lets the rates move. Under a time-varying per-capita birth rate the species count stays geometric and depends on the rate law only through its integral, which under logistic growth converges and leaves a proper limit law rather than merely a finite late-time mean (Corollary 9.3). In a model of Pimentel's competition experiment the state at which the chain is equally likely to step up as down is available in closed form and is unstable ((9.21)), which is a different thing from a process approaching an equilibrium. Two further models are carried only as far as simulation. The chapter's general observation is worth repeating, because it is not the obvious one: what costs the generating function is not feedback as such but *continuous bending* of a rate. A process may act on its own circumstances as much as it likes and stay solvable, provided what survives the interaction is a random time or a random count rather than a moving rate.

Ch.	What it establishes	What it leaves
2	Objects and methods; survival of founding cohorts near criticality; a family of absorption models solved	The absorption models are not used by the modelling chapters
3	$A(p)$ exists, equals a Koenigs value, has no elementary formula in z , and is computable across $(0, \frac{1}{2})$	Whether $p \mapsto A(p)$ has a closed form
4	The birth–death–catastrophe cell: fates, moments, mean release, burst moments, all through $I(t)$	Rates unmeasured for <i>Y. pestis</i> ; no data confrontation
5	Geometric load at every time; quasi-stationary law; burst-size law identical to it, with a structural proof	The chained process for $\mu > 0$; whether the telescoping criterion is necessary
6	An exact burst-aware renewal model; the standard model as its projection; non-identifiability; the flooding criterion $L > 1$	Where <i>Y. pestis</i> sits relative to $L = 1$; multiplicity through the population layer
7	Exact hypergeometric no-catastrophe probability for two types with one global catastrophe	Killed second moments, which an eclipse-aware renewal model needs
8	Depletable defences: certainty threshold, $\mathcal{R}(\vartheta)$, the reversal at $L = 1$, and the budget ϑ^*	Population-level consequences of a threshold clearance
9	Distribution under a moving birth rate; logistic limit law; an unstable moving critical point	Four of the models are specifications or simulations only

Table 10.1: What each chapter settles, and what it hands on. The right-hand column is drawn from the chapters' own statements of scope, not added here.

10.2 The four questions, answered

10.2.1 Why *Y. pestis* switches phase

Two arguments in this thesis reach the switch, by different routes, and they agree on its shape if not yet on its position. The first is the quadrant argument of [section 10.3](#): with three environments ordered by hostility, the expected return on accepting phagocytosis falls linearly as the neutrophil fraction rises, crosses the constant return on resisting it, and the crossing sits strictly inside $(0, 1)$. The second is the removal budget of §8.5.4: while the killer-cell density is roughly constant a cell releasing its whole load at once pays one toll where a budding cell would pay many, and once a rise in that density pushes the toll past $\vartheta^* = \log V_\infty / \log a$ no intracellular strategy remains supercritical. A third strand supports both without deciding either. Just above criticality the size of a founding cohort weighs very heavily on whether a lineage survives at all (Section 2.A), and the intracellular phase is exactly a device for arriving in numbers.

The common prediction is worth isolating, because it is the part that could be wrong in an interesting way. Both arguments key the switch to *how much of the killer field is present*, not to a clock: to the fraction of phagocytes that are neutrophils in one case, to the number of agents removed per delivery in the other. Neither predicts a fixed interval after inoculation. The histology does place the switch alongside the neutrophil response [\[2, 3\]](#), which is consistent with both accounts and discriminates between neither, and whether the neutrophils drive the switch or merely arrive with it is not settled by those studies.

Three things cannot be claimed. The two thresholds have not been shown to be consistent — whether the neutrophil fraction that crosses p_N^* is the one that carries the toll past ϑ^* is an open problem recorded in §8.6.2 and repeated in [section 10.4](#). Neither threshold has been put against data. And neither argument excludes the plainer explanations: that macrophages are being used as transport to the draining lymph nodes, or that the early pro-phagocytic phase is a hangover from life inside the flea. What the mathematics supplies is not a proof that the strategy is optimal but a statement of what would have to be measured for the question to be settled.

10.2.2 Whether bursting pays

It does, but “bursting pays” turns out not to be one claim. Four comparisons run through Chapter 6 and Chapter 8, and they do not agree with one another, which is the finding rather than a defect.

At fixed total output per cell and linear clearance, replacing a constant drip by a single burst leaves R_0 exactly unchanged: the schedule is invisible to the reproduction number. At the same fixed mean, the schedule is nevertheless not invisible to establishment, and bursting strictly lowers the extinction probability of a small inoculum precisely when $L = a(1 - b)$ exceeds one. Against a defence with a fixed removal budget rather than a clearance rate, that criterion reverses exactly, and bursting wins when $L < 1$. A fourth comparison, holding the release law fixed and asking how the same total fares when split across more deliveries, favours concentration unconditionally, because a toll paid per delivery is paid once by a bursting cell and n times by a budding one.

So the honest answer is that a depletable defence supplies a genuine selection gradient towards fewer and larger deliveries, and that whether one particular bursting law beats one particular budding law at matched mean depends on whether the defence is a rate or a budget, with L the quantity that decides. What remains out of reach is placing *Y. pestis* on that scale. Doing so needs λ , μ and δ for the macrophage phase, and δ is fixed by the exit mechanism, which is exactly the thing the primary literature still treats as unresolved — apoptosis, pyroptosis and necroptosis are all reported [\[4, 5, 3\]](#). The two parameter sets available in the right form

belong to *F. tularensis* [6] and to *B. anthracis* [7], and they describe cells at opposite extremes of what the model allows — near-certain rupture releasing hundreds in the one case, near-certain clearance with minimal release in the other. Neither is *Y. pestis*.

10.2.3 An intracellular replacement for the standard model

Yes, and it is more than a substitution of one rate for another. The standard model gives an infected cell two constants, a release rate and a removal rate, and for a budding virus that is an exact description rather than an approximation. For a lytic pathogen it is wrong in mechanism, because release and death are not independent events but the same event. The renewal system of Chapter 6 replaces the two constants with two kernels derived from the intracellular process, is exact in the mean, and returns the standard model as a projection of itself, valid one exponential phase at a time. Three consequences bear directly on fitting practice: acute-phase and set-point estimates of the release rate should differ systematically and by a bounded amount; the eclipse emerges from the intracellular dynamics with no extra parameter, so a model carrying a fitted one may be counting the same delay twice; and because the population fit is not identifiable for the three intracellular rates, it is worth doing only alongside a single-cell companion measurement. The single-cell predictions that would supply that companion are listed in §5.12.1, and are sharp enough to be wrong.

10.2.4 Rates that move

Partially, and the partial answer has a clean edge to it. Three results in Chapter 9 carry closed forms — the geometric law under a moving birth rate, the logistic-speciation limit law, and the unstable moving critical point of the Pimentel chain — and the remaining models are specifications or simulations that outrun the machinery of the earlier chapters. What that collection shows is where the machinery gives out, and the boundary is not the expected one. It is not feedback that costs the generating function. A process may consume a capacity, harden a competitor or infect its neighbours and still be solvable, provided the interaction leaves behind a random time or a random count rather than a rate that goes on bending. That is a modest observation, but it is a usable one: it says which of the models in that chapter are worth attacking analytically and which should be left to simulation.

10.3 The quadrant argument

The argument here is the crudest in the thesis and, for that reason, the one whose assumptions are easiest to see. It sorts the bacterium's options into a two-by-two — accept or resist phagocytosis, before or after the neutrophil influx — and asks which cell of that grid offers the better return. The claim is that the best cell moves along the diagonal as the influx proceeds, and that it does so at a threshold which can be written down.

10.3.1 Three environments, one ordering

Simplify the intracellular options to macrophages and neutrophils, setting aside dendritic cells, eosinophils, basophils and mast cells, which are no doubt relevant in reality. Macrophages are generalist phagocytes; neutrophils are specialist killers, and it is mainly neutrophils that are recruited in numbers once a bacterial infection of this kind is recognised. Treat replication in each of the three environments — inside a macrophage, inside a neutrophil, and extracellular — as a simple birth–death process with rates (β_i, μ_i) for $i \in \{M, N, E\}$. The difference in hostility then sits in the death rate, and

$$\mu_N > \mu_E > \mu_M. \quad (10.1)$$

The outer inequalities are the substance of the model, and each is an inference from the strategy itself rather than an independent measurement. That *Y. pestis* evolved to exploit the macrophage interior at all is the reason for taking $\mu_M < \mu_E$; if there were no advantage inside, there would be no reason to go in. That it resists phagocytosis once neutrophils arrive in force is the reason for taking $\mu_E < \mu_N$; if the neutrophil interior were the better place, resisting entry to it would be perverse. The argument is therefore not a derivation of the strategy from first principles. It is a consistency check: it asks whether one ordering of three death rates can account for both halves of the observed behaviour at once.

Take the birth rates equal,

$$\beta_M = \beta_E = \beta_N = \beta, \quad (10.2)$$

which is certainly false in detail — nutrient and energy supply differ inside a cell and outside it — but which keeps the comparison to one axis. What matters to the comparison is the net rate $\beta_i - \mu_i$ in any case, and by [equations \(10.1\) and \(10.2\)](#),

$$\beta_M - \mu_M > \beta_E - \mu_E > \beta_N - \mu_N. \quad (10.3)$$

The macrophage is the best environment available. The bacterium cannot, however, choose it: the assumption here is that *Y. pestis* can decide whether to permit phagocytosis but cannot decide which phagocyte performs it.

Remark 10.1 (What the argument does not consider). There are other reasons *Y. pestis* might favour macrophage uptake early. The infected macrophage may serve chiefly as transport to the draining lymph nodes, where the food supply is better; and the pro-phagocytic phase may be a lingering hangover from life inside the flea, where the temperature cue for the second phase is absent. Neither possibility is excluded by anything below. The argument explores whether a more fundamental strategic advantage is available in addition, not instead.

Also folded into a constant here is the mechanism of the killing itself. Leucocytes kill from a store of granules, and §8.3 shows that when the store is depletable the effective death rate falls as a cohort spends it — so μ_M and μ_N are not constants but declining functions of how much killing has already been done. Taking them fixed is the approximation that makes the present argument a two-line calculation, and [section 10.4](#) records it as the place where the argument is weakest.

10.3.2 Where the two strategies cross

Following pathogen recognition the body delivers neutrophils to the site, so that the composition of the phagocyte population shifts. Write

$$p_M = \frac{\#M}{\#M + \#N}, \quad p_N = \frac{\#N}{\#M + \#N} = 1 - p_M, \quad (10.4)$$

for the probability that an uptake event is performed by each type, and take p_N to rise at the influx. A bacterium that permits phagocytosis therefore draws its environment at random from the two, while one that resists it takes the extracellular rate with certainty:

$$R_{\text{pro}} = p_M(\beta_M - \mu_M) + p_N(\beta_N - \mu_N), \quad (10.5)$$

$$R_{\text{anti}} = \beta_E - \mu_E. \quad (10.6)$$

Eliminating p_M makes the dependence on the influx explicit,

$$R_{\text{pro}} = (\beta_M - \mu_M) + p_N [(\beta_N - \mu_N) - (\beta_M - \mu_M)], \quad (10.7)$$

so that R_{pro} falls linearly in p_N , from $\beta_M - \mu_M$ at $p_N = 0$ to $\beta_N - \mu_N$ at $p_N = 1$, while R_{anti} does not move. The two therefore cross exactly once. Resisting phagocytosis becomes the better strategy when $R_{\text{anti}} > R_{\text{pro}}$, which by (10.7) — and remembering that the bracket is negative, so that the inequality reverses on division — occurs when

$$p_N > \frac{(\beta_E - \mu_E) - (\beta_M - \mu_M)}{(\beta_N - \mu_N) - (\beta_M - \mu_M)}, \quad (10.8)$$

and with the birth rates equal this reduces to

$$p_N > p_N^* := \frac{\mu_E - \mu_M}{\mu_N - \mu_M} \quad (10.9)$$

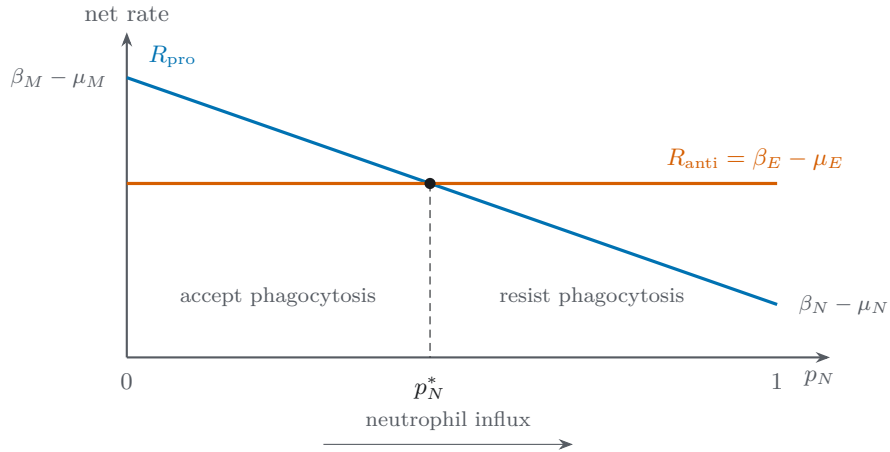


Figure 10.1: The crossing. The expected net replication rate under a pro-phagocytic strategy falls linearly in the neutrophil fraction p_N , because a larger fraction of uptake events delivers the bacterium into the more hostile of the two cell types; the anti-phagocytic rate does not depend on p_N at all. The single crossing is p_N^* of (10.9), and it lies strictly between 0 and 1 exactly when the ordering (10.1) holds.

10.3.3 What the threshold says

(10.9) has a reading that makes the ordering assumption do visible work rather than sit in the background. The threshold is the position of the extracellular death rate on the interval running from the macrophage rate to the neutrophil rate: if the outside world is a quarter of the way from macrophage to neutrophil in hostility, the switch comes when a quarter of the phagocytes on the scene are neutrophils. That reading also shows what the ordering buys, since p_N^* lies in $(0, 1)$ if and only if (10.1) holds. Were the extracellular environment kinder than the macrophage interior, the threshold would be negative and resistance would be right from the first moment; were it harsher than the neutrophil interior, the threshold would exceed one and the bacterium should accept phagocytosis however many neutrophils arrive. A biphasic strategy is optimal only in the intermediate case, and the intermediate case is precisely the ordering inferred from the strategy. The argument is a consistency check that passes, and it should be read as no stronger than that.

Two things follow that are not merely restatements. The first is that the switch is keyed to a composition and not to an elapsed time, so anything that changes the arriving mixture should move it. Neutrophil depletion should delay the switch or remove it; accelerated recruitment should bring it forward. That is the same shape of prediction as the removal budget ϑ^* of §8.5.4, reached from a different model with different machinery, and the two accounts have not

been reconciled. The second is that p_N^* is written entirely in death rates, which are in principle measurable: intracellular survival of *Y. pestis* in macrophages and in neutrophils, and survival in the extracellular medium, are three quantities of the same kind, and the argument stands or falls on whether the middle one lies between the other two.

Finally, the argument as set out is indifferent to *when* the influx happens. That the bacterium's residence in macrophages may itself delay neutrophil recruitment, by hiding the infection, is a real effect and would show up as t_N depending on the strategy chosen. It does not enter here, because whether uptake is advantageous at a given moment depends on the mixture present at that moment and not on the history that produced it.

10.4 Limitations and open problems

Nothing in this thesis has been fitted to plague data. The rates the models need for the intracellular phase of *Y. pestis* do not exist in the literature in the form required, and the two published parameter sets used for illustration in Chapter 5 belong to other organisms. Everything said about *Y. pestis* is therefore a hypothesis compatible with the cited intracellular literature, not an inference this work can check.

Four assumptions carry the closed forms and none is innocuous. Birth is linear, so the conditional load has no intracellular carrying capacity; the defence is that the rupture hazard is itself proportional to load and so regulates it, but the sensitivity of the burst statistics to a logistic birth rate has not been checked. The rupture hazard is exactly δn — not δn^2 , not a critical load, not a maturation-timed lysis — and this is the strongest assumption anywhere in the thesis, because the burst law and its telescoping proof both depend on it. Release is instantaneous and complete, which is what makes the burst a single random variable; an assay that saw graded release would refute the premise before reaching any of the theory. And the rates are constant in time, which Chapter 9 relaxes only in models that no longer close.

The quadrant argument of [section 10.3](#) is the weakest piece of mathematics in the thesis, and it is kept because it is transparent rather than because it is strong. It treats μ_M and μ_N as constants at exactly the point where Chapter 8 shows them not to be: killing from a depletable granule store means the death rate falls as a cohort spends the store, so the comparison between environments should depend on cohort size, and in this argument it does not. It also compares instantaneous rates rather than probabilities of establishment, which is the quantity that actually decides the fate of a small founding population.

Two structural gaps remain between the chapters. There is no host-to-host layer: the released particles are counted and then forgotten, so the population-level consequences of a threshold clearance are not derived anywhere. And multiplicity of infection, which the single-cell theory handles cleanly, has not been carried through the population layer, even though the kernels for it are available and add no fitted parameters.

Of the open problems recorded across the chapters, four are the ones worth taking first.

1. **Reconcile the two thresholds.** Whether the neutrophil fraction that crosses p_N^* in (10.9) is the same event as the toll crossing ϑ^* in Corollary 8.8. Two accounts of one transition, in different currencies, is one account too many.
2. **Two-type killed second moments.** These are what an eclipse-aware renewal model needs, and they are the principal technical obstacle between Chapter 6 and Chapter 7. The one-type escape was to differentiate the Riccati equation; the two-type route should be to differentiate the backward system. It should not be assumed easy.
3. **Locate *Y. pestis* relative to $L = 1$.** This is the question that would turn the bursting comparison from a criterion into a statement about the organism, and it needs δ , which needs the exit mechanism.

4. **The chained process at $\mu > 0$.** The joint laws of rupture times and sizes are known in closed form only when internal extinction is impossible, and the decomposition fails as soon as a lineage has two ways to stop rather than one.

10.5 Closing

Set against one switch, what this thesis can offer is two thresholds and a parameter. The thresholds are p_N^* , the neutrophil fraction at which being eaten stops paying, and ϑ^* , the removal budget beyond which no intracellular strategy stays supercritical; the parameter is $L = a(1 - b)$, which decides whether releasing everything at once is a good bet or a bad one and changes its answer according to whether the defence facing the release is a rate or a budget. None of the three is measured. All three are written in quantities that could be.

That is a narrower return than the question seemed to promise, and the reason is worth naming, because it is also the pattern that recurred throughout. So much of the biology turns on arriving in numbers rather than on replicating quickly. A near-critical population is almost indifferent to its growth rate and acutely sensitive to how many founders it started with; depletable defences reward the cohort that spends them together; and the whole intracellular phase reads, from far enough back, as a device for converting a slow advantage into a large one at a single instant. The mathematics of that conversion is what the middle chapters are about, and the burst-size law is where it is at its most compact: a geometric distribution whose ratio is the larger root of a quadratic in three rates.

Chapter 9 came from a different appetite, and it marks plainly which of its parts are results and which are hunches. The suspicion that drove it — that tension accumulates and releases in cycles across scales, and that the release is where new form appears — remains a suspicion. What it produced that is defensible is narrower and worth having: the observation that a process may act on its own circumstances and stay solvable, so long as what crosses the join is a random time rather than a bending rate. *Y. pestis* is a good illustration of that principle, since the whole of its intracellular phase reaches the extracellular one through a single number, the size of the burst.

What would settle the questions this thesis leaves open is not more theory. It is three death rates and a rupture rate, measured for one organism in one cell type: time-lapse imaging of infected macrophages held to rupture, with the moment of release recorded and the payload counted. The predictions are sharp enough to be wrong, which is the most that a model of this kind should be asked to be.

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